

An Introduction to E-Commerce

with a Focus on Machine Learning, Deep Learning, and Artificial Intelligence

Lecture 14: Capstone: Enterprise-Grade AI E-Commerce Architecture

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Session plan (90 minutes)

- ① Capstone scenario
- ② Purpose of the capstone
- ③ The company context
- ④ Capstone design question
- ⑤ Expected architectural scope
- ⑥ Final reflection

Capstone scenario

- Scenario: an end-to-end "AI commerce + Odoo" reference architecture for a single company.
- Goal: demonstrate enterprise architecture alignment, ERP integration, and measurable AI value with operational readiness.
- Business model: hybrid B2C + B2B commerce.
- Commerce front-end: compare Odoo Website/eCommerce with a separate headless storefront integrated to Odoo.
- AI stack: combine classical ML, deep learning, and LLM-based service automation where justified.

Purpose of the capstone

- The capstone is the point at which isolated topics become a system.
- Earlier chapters covered platforms, data foundations, enterprise architecture, integration, ERP connectivity, recommender systems, pricing, fraud, LLM support, MLOps, security, and governance.
- In real enterprises, however, these elements do not operate as separate course modules.

The company context

- increase online conversion and average order value;
- improve cross-sell and upsell performance;
- reduce fraud losses and operational false positives;
- shorten support resolution time while maintaining policy compliance;
- improve inventory visibility and order orchestration across channels;
- establish a governed AI operating model that can scale beyond a single experiment.

Capstone design question

- Students must answer one central question: This question is intentionally broader than a software implementation exercise.
- It requires decisions about system boundaries, ownership, integration patterns, data products, operating model, risk controls, and business metrics.

Expected architectural scope

- customer and business-facing channels;
- commerce application services;
- enterprise systems such as ERP, CRM, WMS, and finance processes;
- integration patterns across synchronous and asynchronous flows;
- data ingestion, storage, analytics, and feature computation layers;
- ML and LLM services for prioritized use cases;

Business capabilities and value streams

- product and catalog management;
- customer identity and account management;
- search, browse, and personalization;
- cart, checkout, payment, and order capture;
- inventory visibility and fulfillment orchestration;
- returns, refunds, and service case management;

Reference solution shape

- **Channel layer:** a branded web storefront for B2C;; a self-service portal for B2B customers with negotiated pricing and account hierarchies;
- **Commerce services layer:** catalog and content services;; pricing and promotion services;
- **Enterprise systems layer:** sales order management;; inventory and warehouse operations;

- **Synchronous interactions:** product detail retrieval;; price and promotion lookup;
- **Asynchronous interactions:** order-created and payment-settled events;; inventory updates;
- **Integration design decisions:** which systems publish canonical events;; how idempotency and replay are handled;

Data and analytics architecture

- behavioral event collection from channels;
- operational data from commerce and Odoo processes;
- curated analytical models for orders, customers, products, and inventory;
- experiment and campaign measurement tables;
- feature pipelines for ML use cases;
- dashboards and semantic metrics for business teams.

- **Recommended initial use cases:** recommender systems for personalized ranking, related products, and bundle suggestions;; pricing and promotion analytics for elasticity-aware offers and margin protection;
- **Use-case selection logic:** expected business value;; availability and quality of data;

Target operating model

- product management to define business objectives and backlog priorities;
- commerce engineering to own channel and transaction experiences;
- enterprise applications teams to own Odoo processes and integrations;
- data engineering to own data pipelines, contracts, and curated datasets;
- ML engineering or applied science to own models, experimentation, and monitoring;
- security, privacy, and compliance functions to govern sensitive use cases;

Security, privacy, and responsible AI controls

- identity and access control across customer, partner, and staff roles;
- payment and transaction security;
- encryption, secret handling, and audit logging;
- data minimization and retention;
- model and LLM governance for high-impact use cases;
- human review and override paths;

MLOps and production readiness

- reproducible training and evaluation workflows;
- model registration and release approval;
- monitoring for technical, model, and business metrics;
- retraining or rollback triggers;
- feature consistency controls between training and serving;
- logging and traceability for audits and incident analysis.

- **Business value:** conversion rate;; average order value;
- **Technical fitness:** architectural coherence;; correctness of system boundaries;
- **Operational credibility:** Can this design actually be implemented by a real organization?; Are roles and ownership explicit?

- **Optional technical annexes:** sample API contracts;; event schemas;

Suggested delivery sequence

- define the business strategy, operating context, and priority value streams;
- map business capabilities and pain points in the current state;
- define the target architecture and key system boundaries;
- select AI use cases and justify them with measurable value hypotheses;
- specify integration, data, and governance mechanisms;
- define rollout phases, metrics, and risks;

Common failure modes in student capstones

- treating every system as equally important rather than identifying the architectural core;
- proposing AI use cases without sufficient data, ownership, or evaluation logic;
- ignoring ERP and operational process realities in favor of front-end innovation;
- omitting security, privacy, and governance until the last page;
- drawing integrations without clarifying system-of-record boundaries;
- assuming that good offline model performance automatically means business value.

Worked example: hybrid headless commerce with Odoo backbone

- the storefront delivers flexible customer experience and experimentation velocity;
- pricing, promotion, and catalog services expose reusable APIs to both channels;
- Odoo handles order orchestration, inventory, procurement, invoicing, and finance synchronization;
- an event bus distributes order, payment, inventory, and behavior events;
- a data platform supports experimentation, recommender training, fraud analytics, and management reporting;
- LLM-based support tools assist agents but do not execute sensitive actions without policy controls.

- The purpose of the capstone is not to prove that one tool or architecture style is universally best.
- It is to show that enterprise e-commerce is a system of systems.
- Channels, enterprise operations, data products, AI services, controls, and organizational responsibilities must align if the business wants durable value.